

II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

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Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

How to beat the curse of dimensions?

Probability distribution of factored systems with states $\times_{k \in [d]} [m_k]$ has

$$\left(\prod_{k \in [d]} m_k \right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: [Tensor Network Decompositions](#)

Independencies of Random Variables

Decompositions of Probability Tensors correspond with independencies of (hidden) random variables.

Example: Being at a dentist

Add a variable **Cloud**, denoting the weather outside the dentists lab.

- ▶ This adds an additional axis to $\mathbb{P}$, thus the number of coordinates increases by a factor of 2.
- ▶ But: Intuitively, knowing **Cloud** should not affect the probability of having a cavity, so why shall we care?

Independence of Cloud to the other Variables

After showing that cavity, catch and toothache are independent of cloud, we do not have to consider cloud any more.

Formal definition of Independencies

Definition (Independence)

Given a joint distribution of variables X_0 and X_1 , we say that X_0 is independent from X_1 if for any values x_{X_0}, x_{X_1} we have

$$\mathbb{P}[X_0 = x_0, X_1 = x_1] = \mathbb{P}[X_0 = x_0] \cdot \mathbb{P}[X_1 = x_1] .$$

Contraction criterion for independence

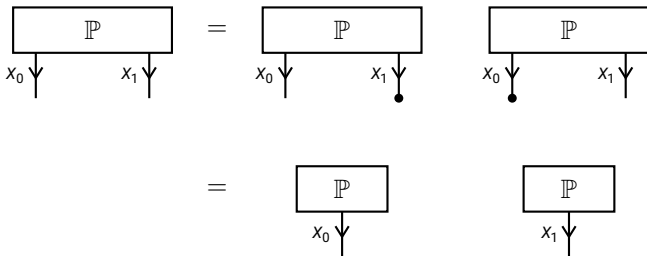
Theorem

Given a probability distribution $\mathbb{P}$, X_0 is independent from X_1 , if and only if

$$\mathbb{P}[X_0, X_1] = \langle \mathbb{P} \rangle_{[X_0]} \otimes \langle \mathbb{P} \rangle_{[X_1]} .$$

Decomposition into Marginal Probabilities

Independence allows the decomposition into



Exponential to linear storage demand

Instead of storing $m_0 \cdot m_1$ coordinates, we can store $\mathbb{P}$ in the case of independency with $m_0 + m_1$ demand.

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- ▶ **Conditional independence**
- ▶ Decompositions given independence
- ▶ Bayesian networks
- ▶ Markov networks

Assumption of independence

Independence is a very restrictive assumption, corresponding with tensor products being the most "extreme" tensor decomposition.

Example: In the dentist example, no variables are independent from each other.

Are there more common properties corresponding to more generic tensor decompositions?

Formal definition of Conditional Independencies

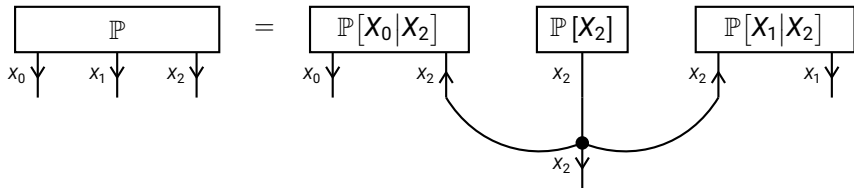
Definition (Conditional Independence)

Given a joint distribution of variables X_0, X_1 and X_2 , we say X_0 is independent from X_1 conditioned on X_2 if for any index tuple (x_0, x_1, x_2)

$$\mathbb{P}[X_0 = x_0, X_1 = x_1 | X_2 = x_2] = \mathbb{P}[X_0 = x_0 | X_2 = x_2] \cdot \mathbb{P}[X_1 = x_1 | X_2 = x_2] .$$

Decomposition given conditional independence

We depict conditional independence by tensor network decompositions:



Example: Being at a dentist

The toothache variable and the cavity variable are independent conditioned on the catch variable.

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- ▶ **Decompositions given independence**
- ▶ Bayesian networks
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Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}[X_{[d]}]$ we have

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{[k]}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: We apply the Bayes theorem to get

$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_0], \mathbb{P}[X_1, \dots, X_{d-1} | X_0] \rangle_{[X_{[d]}]} .$$

and further apply the Bayes theorem iteratively on the conditioned distribution.

Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_k where $k \in [d]$. When X_k is independent of $X_{[k-1]}$ conditioned on X_{k-1} (the Markov Property), then

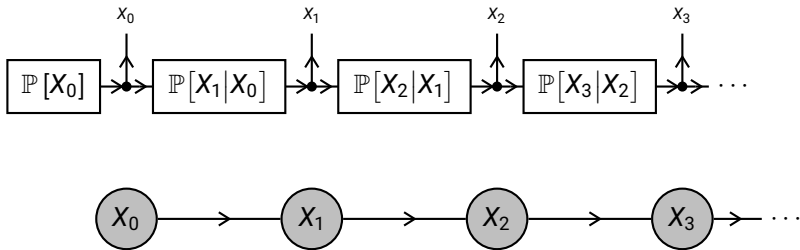
$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_k | X_{k-1}] : k \in [d] \rangle_{[X_{[d]}]} .$$

Proof sketch: We use the chain decomposition and exploit the independence assumptions to get

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{k-1}] \otimes \mathbb{I}[X_{[k-1]}] .$$

Tensor network decompositions of markov chains

We depict this by



Can we find a more generic correspondence between probability decompositions and conditional independence statements?

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- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decompositions given independence
- ▶ **Bayesian networks**
- ▶ Markov networks

Independence assumptions of Bayesian networks

- ▶ We enumerate the variables by $[d]$.
- ▶ To each $k \in [d]$ we choose a set $\text{Pa}(k)$ of **parent nodes** in $[k] = \{0, \dots, k-1\}$ such that

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

Example: Markov chains are the special case of $\text{Pa}(k) = \{k-1\}$.

Factorization into Bayesian networks

Theorem

When for each $k \in [d]$

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

then

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{\text{Pa}(k)}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: As in the proof of the Markov chain factorization theorem, we use the chain factorization and exploit the conditional independences to show that

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \mathbb{I}[X_{[k] \setminus \text{Pa}(k)}] .$$

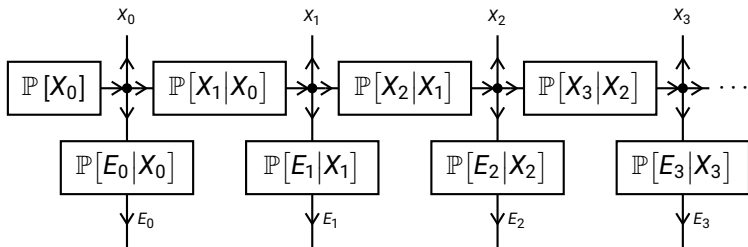
Example: Hidden Markov Models

Hidden Markov Models extend Markov Chains by limited observation E_k of the variables X_k .

The independence assumptions are

- ▶ X_{k+1} is independent of $X_{0:k-1}$ and $E_{0:k}$ conditioned on X_k
- ▶ E_k is independent of all other variables conditioned on X_k

The independence assumptions are exploited in the decomposition



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- ▶ **Markov networks**

Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\{\tau^e[X_e] : e \in \mathcal{E}\}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}[X_{\mathcal{V}}] = \frac{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}]}}{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}} = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}|\emptyset]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}$$

the **partition function** of the Markov Network.

Markov network independencies

Theorem

A positive distribution $\mathbb{P}[X_{\mathcal{V}}]$ has a representation as a Markov network on a hypergraph $\mathcal{G}$, if and only if for any disjoint subsets $\mathcal{U}, \mathcal{W}, \mathcal{I} \subset \mathcal{V}$ for which $\mathcal{I}$ separates $\mathcal{U}$ from $\mathcal{W}$

$$(X_{\mathcal{U}} \perp X_{\mathcal{W}}) \mid X_{\mathcal{I}}.$$

Proof sketch: This is the Hammersley Clifford factorization theorem, which proof is out of scope in this course.